Angelo Rodio



Postdoctoral Researcher in Decentralized Machine Learning

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Italian · born 6 July 1995 in Bari, Italy

Research interests

Decentralized and federated learning over networks: convergence theory, communication design, privacy, and energy efficiency. Methods: convex and non-convex optimization, linear algebra, graph theory, stochastic modeling, and statistical learning theory.

Positions

Sep 2024 – present	**Postdoctoral Researcher**, Linköping University, Sweden Division of Communication Systems, Department of Electrical Engineering. Hosts: Prof. Erik G. Larsson and Assoc. Prof. Zheng Chen.
Jul – Sep 2026	**Visiting Researcher**, Nanyang Technological University (NTU), Singapore College of Computing and Data Science. Host: Assoc. Prof. Chee Wei Tan. Funded by my Vinnova mobility grant (project MADELEINE).
Sep 2020 – Mar 2021	**Telecommunications Engineer**, TIM S.p.A., Rome, Italy Technical consulting, implementation and integration support on a cloud platform.

Education

Apr 2021 – Jul 2024	**PhD in Computer Science**, Inria and Université Côte d'Azur, France Thesis: *Client Heterogeneity in Federated Learning Systems*, defended 3 July 2024. Supervisors: Giovanni Neglia and Alain Jean-Marie. Reviewers: H. Vincent Poor (Princeton University) and Michele Rossi (University of Padova). Funded by a 3IA Côte d'Azur doctoral fellowship.
Sep 2019 – Aug 2020	**MSc in Computer Science (double degree)**, Université Côte d'Azur, France Grade: *Très bien*. Thesis on Wi-Fi fingerprint indoor localization for smartphones, at Inria (supervisors: Arnaud Legout and Chadi Barakat).
Oct 2018 – Sep 2020	**MSc in Telecommunications Engineering**, Politecnico di Bari, Italy Grade: 110/110 *cum laude*. Supervisor: Luigi Alfredo Grieco.
Sep 2014 – Feb 2018	**BSc in Computer Science**, Politecnico di Bari, Italy Grade: 109/110. Thesis on the performance of nanoscale communication systems (supervisors: Giuseppe Piro and Gennaro Boggia).
2026	**Qualification aux fonctions de Maître de conférences**, France National qualification for associate professor positions, CNU Section 27 (Computer Science).

Grants and awards

2026	**Vinnova research grant**, call *Individual mobility within 6G for collaborations with the USA and Singapore*. Project MADELEINE: *Massive Decentralised Learning in the Network Edge* (ref. 2026-01731), SEK 468,003. Sole applicant and project leader; funds a research stay at NTU Singapore and a new LiU–NTU collaboration.
2025	**IEEE Information Theory Society travel grant**, IEEE ITW 2025.
2024	**ACM SIGMETRICS student travel grant**, ACM SIGMETRICS 2024.
2021	**3IA Côte d'Azur doctoral fellowship** (PhD funding).

Publications

Google Scholar: scholar.google.com/citations?user=l-uI1ioAAAAJ. My name in bold.

Journal articles

[4] D. R. Arputharaj, C. Rodriguez, **A. Rodio**, G. Neglia. Carbon-Aware Client and Time Scheduling for Green Federated Learning. *Performance Evaluation*, art. 102597, 2026. doi:10.1016/j.peva.2026.102597

[3] C. Xu, C. Kaplan, **A. Rodio**, T. Si Salem, G. Neglia. Federated Learning for Collaborative Inference Systems: The Case of Early Exit Networks. *Performance Evaluation*, vol. 171, art. 102538, 2026. doi:10.1016/j.peva.2025.102538

[2] **A. Rodio**, F. Faticanti, O. Marfoq, G. Neglia, E. Leonardi. Federated Learning Under Heterogeneous and Correlated Client Availability. *IEEE/ACM Transactions on Networking*, 32(2):1451–1460, 2024. doi:10.1109/TNET.2023.3324257

Conference papers

[9] **A. Rodio**, G. Neglia, Z. Chen, E. G. Larsson. A Unified Convergence Analysis for Semi-Decentralized Learning: Sampled-to-Sampled vs. Sampled-to-All Communication. *AAAI Conference on Artificial Intelligence (AAAI)*, 2026. **Oral presentation.** doi:10.1609/aaai.v40i30.39705

[8] D. Pérez Herrera, **A. Rodio**, Z. Chen, E. G. Larsson. Power Allocation for Average Consensus in Wireless Networks with Link Failures. *European Wireless Conference (EW)*, pp. 172–177, 2025.

[7] D. R. Arputharaj, C. Rodriguez, **A. Rodio**, G. Neglia. Green Federated Learning via Carbon-Aware Client and Time Slot Scheduling. *IEEE MASCOTS*, 2025. doi:10.1109/MASCOTS67699.2025.11283314

[6] **A. Rodio**, Z. Chen, E. G. Larsson. Optimizing Privacy-Utility Trade-off in Decentralized Learning with Generalized Correlated Noise. *IEEE Information Theory Workshop (ITW)*, 2025. doi:10.1109/ITW62417.2025.11240322

[5] **A. Rodio**, G. Neglia. FedStale: Leveraging Stale Updates in Federated Learning. *European Conference on Artificial Intelligence (ECAI)*, pp. 3071–3078, 2024. doi:10.3233/FAIA240849

[4] **A. Rodio**, G. Neglia, F. Busacca, S. Mangione, S. Palazzo, F. Restuccia, I. Tinnirello. Federated Learning with Packet Losses. *International Symposium on Wireless Personal Multimedia Communications (WPMC)*, 2023. doi:10.1109/WPMC59531.2023.10338845

[3] **A. Rodio**, F. Faticanti, O. Marfoq, G. Neglia, E. Leonardi. Federated Learning under Heterogeneous and Correlated Client Availability. *IEEE INFOCOM*, 2023. doi:10.1109/INFOCOM53939.2023.10228876

[2] **A. Rodio**. The Many Facets of Variance Reduction in Federated Learning. *ACM SIGMETRICS Performance Evaluation Review* 52(4):11–12, 2025 (poster at ACM SIGMETRICS 2024). doi:10.1145/3725536.3725542

Under review

[2] D. Pérez Herrera, **A. Rodio**, Z. Chen, E. G. Larsson. Heterogeneity-Aware Topology and Mixing Matrix Design for Decentralized Learning. Submitted to *IEEE Transactions on Signal and Information Processing over Networks*.

PhD thesis

[1] **A. Rodio**. Client Heterogeneity in Federated Learning Systems. PhD thesis, Université Côte d'Azur, 2024. HAL: tel-04685040

Teaching

Master's level, in English, in Sweden and France: 79 hours in total.

Fall 2026 — **Distributed Information Processing and Machine Learning (TSKS38)**, Linköping University
28 h: 12 h of tutorials and 16 h of labs (with Adrian Edin).

Fall 2025 — **Distributed Information Processing and Machine Learning (TSKS38)**, Linköping University
12 h of tutorials; prepared the problems and solutions, and graded the exams.

2022 – 2024 — **Federated Learning and Data Privacy**, Université Côte d'Azur (M2 DS4H)
12 h of labs per year (2022–23 and 2023–24, including 3 h co-taught with C. Xu).

2021 – 2022 — **Optimization for Machine Learning**, Université Côte d'Azur (M1 DS4H)
15 h of labs (including 9 h co-taught with O. Marfoq).

Supervision

2026	**Mehran Mamivand**, MSc thesis, Linköping University. *Energy-Efficient Semi-Decentralized Federated Learning via Cooperative D2D Model Aggregation*.
2026	**Camilla Samuelsson**, MSc thesis, Linköping University, carried out at the Swedish Defence Research Agency (FOI). *Set Reconciliation Algorithms in Tactical Networks*.
2024	**Albachiara Bellaroba**, MSc thesis, Sapienza University of Rome, Italy.
2023	**Mohamed Salah Jebali**, MSc thesis, University of Florence, Italy.
2023	**Maysae Hmamouchi**, MSc research project, Université Côte d'Azur, France.

Service

Session chair	AAAI Conference on Artificial Intelligence (AAAI 2026); IEEE Information Theory Workshop (ITW 2025).
Reviewer	*Journals:* IEEE Journal on Selected Areas in Communications; IEEE Transactions on Automatic Control; IEEE Transactions on Mobile Computing; Computer Networks. *Conferences:* AAAI, UAI, IEEE INFOCOM, IEEE GLOBECOM, IEEE ICC, IEEE ICASSP.
Evaluation	Evaluator of Master's final projects, MSc Data Science and Artificial Intelligence, Université Côte d'Azur, 2023.

Skills and languages

Programming	Python, C/C++, Java; Linux, Git, Jupyter notebooks; reproducible research code.
Languages	Italian (native), English (C1), French (B2).

References

Prof. Erik G. Larsson
Professor and Head of the Division of Communication Systems, Linköping University, Sweden
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Dr. Giovanni Neglia
Research Director, Inria, Head of the NEO team, 3IA Côte d'Azur Chair, France
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